Chapter 4 — Functions, Scope and Closures

Functions let you name behaviour, reuse it, pass it to other code and return it from other functions. Mastering them means mastering **scope** (which variables code can see) and **closures** (functions that remember the variables around them) — the ideas behind private state, event handlers (functions run when something happens, like a click), React hooks (functions like `useState` that give components memory between screen updates) and many interview questions.

What you'll learn: defining/calling functions (parameters vs arguments, `return`, missing/extra arguments, passing values); declarations vs expressions and hoisting; arrow functions (every form and difference); defaults (evaluated at call time), rest, `arguments`, spread, options objects; global/function/block scope, lexical scoping, the scope chain, shadowing; hoisting in full; closures (counters, module pattern, factories, `once`, `memoize`, the loop puzzle); higher-order functions, callbacks, composition, currying; IIFEs, recursion, pure functions.

Code: `code/` — every "Output:" is real Node 24 output. `this` has its own chapter: Chapter 9 — `this`, Prototypes and Classes.

4.1 What a function is

What it is

A **function** is a named, reusable block of code. You **define** it once and **call** (**invoke**) it as often as you like. It takes inputs (**parameters**) and can hand back one output (a **return value**).

Why it matters

- **Reuse** — write logic once.
- **Readability** — `calculateShipping(order)` explains itself.
- **Modularity** — small functions with clear inputs/outputs are easy to test (Chapter 16) and change safely.
- **Abstraction** — callers need *what*, not *how*.

How it works — anatomy of a function declaration

```
function functionName(parameter1, parameter2) {
  // body: statements that run when the function is called
  return someValue;   // optional
}
```

Part	Meaning
`function`	keyword that starts a function **declaration**
`functionName`	the name you'll call it by; by convention a verb in camelCase (`getUser` , `isValid`)
`(parameter1, parameter2)`	**parameters**: local variables that receive the inputs; zero or more
`{ … }`	the **body**
`return value`	ends the function and sends `value` back to the caller

Calling `functionName(a, b)`:

1. **Argument** expressions are evaluated left to right.
2. A new **execution context** (private workspace with its own locals) is pushed onto the **call stack** (the engine's list of running functions).
3. Parameters are bound to the argument values in order.
4. The body runs until `return` or its closing brace.
5. The context is popped; the call evaluates to the returned value, or `undefined`.

Parameter vs argument: the *parameter* is the name in the definition (`name`); the *argument* is the value supplied in a call (`"Alice"`).

Example

```
// code/01_function_basics.js (abridged — each output line labels what it shows)
function square(number) {              // return sends a value back to the caller
  return number * number;
}
function describe(a, b) { return `a=${a}, b=${b}`; }
console.log(describe(1), "|", describe(1, 2, 3));   // missing -> undefined, extra ignored

function tryToChange(num, obj) {
  num = 100;                 // changes only the local copy
  obj.changed = true;        // changes the shared object
  obj = { replaced: true };  // rebinding the local name doesn't affect the caller
}
let n = 1; const o = {};
tryToChange(n, o);
console.log("after call: n =", n, "| o =", o);
```

Output (full run):

```
Hello, world!
Hello, world!
Hello, Alice!
square(4) = 16 | square(square(2)) = 16
function without return gives: undefined
negative non-negative
a=1, b=undefined | a=1, b=2
isEven(4): true | isEven(5): false
typeof isEven: function | name: isEven | length: 2
square without () is the function itself: function | with (): 9
after call: n = 1 | o = { changed: true }
```

1. Results can feed other calls: `square(square(2))`.
2. **No `return` → `undefined`**; `return` exits immediately.
3. **Argument counts are never checked:** missing → `undefined`, extras ignored (still reachable, §4.4).
4. **Functions are values (first-class)**: stored, passed, given properties. `typeof` → `"function"`; `.name` is the name; `.length` the number of declared parameters.
5. **`square` vs `square()`**: the function itself vs calling it. Passing `handleClick()` where a function is expected passes its *result*.
6. **Pass by value, where an object's value is a reference** (Chapter 2 §2.4) — "call by sharing": the function can mutate the caller's object, but reassigning the parameter only changes the local name.

Common mistakes

- **Forgetting `return`** → caller gets `undefined`.
- **Return value on the next line** → `undefined` (ASI, Chapter 1 §1.6).
- **Calling instead of passing**: `addEventListener("click", save())` runs `save` immediately.

TL;DR

Define once, call many times; missing arguments and missing `return` give `undefined`; functions are values.

4.2 Declarations vs expressions (and hoisting)

What it is

	Function declaration	**Function expression**
Looks like	`function add(a, b) { … }` as a statement on its own	`const add = function (a, b) { … };` — a function used as a value
Name	required	optional (anonymous, or named for internal use)
Hoisting	**fully hoisted** — callable before its line	follows the variable: `const` / `let` → TDZ (Temporal Dead Zone) error, `var` → `undefined` (so calling it gives `TypeError`)
Typical use	top-level helpers, named utilities	callbacks (functions handed to other code to call later), conditional definitions, values stored in objects

The **arrow function** (`const add = (a, b) => a + b;`, §4.3) is also an expression.

Example

```
// code/02_declarations_expressions_hoisting.js
console.log("declared later, called early:", add(2, 3));
function add(a, b) { return a + b; }

try {
  multiply(2, 3);
} catch (e) {
  console.log("const expression called early ->", e.name + ":", e.message);
}
const multiply = function (a, b) { return a * b; };

try {
  divide(6, 3);
} catch (e) {
  console.log("var expression called early ->", e.name + ":", e.message);
}
var divide = function (a, b) { return a / b; };

const factorial = function fact(n) { return n <= 1 ? 1 : n * fact(n - 1); };
console.log("factorial(5):", factorial(5), "| typeof fact outside:", typeof fact);

const anon = function () {};
const arrowFn = () => {};
console.log("inferred names:", anon.name, arrowFn.name);

(function () {
  "use strict";
  if (true) { function insideBlock() { return "hi"; } }
  console.log("strict: block function visible outside block?", typeof insideBlock);
})();
```

Output:

```
declared later, called early: 5
const expression called early -> ReferenceError: Cannot access 'multiply' before initialization
var expression called early -> TypeError: divide is not a function
factorial(5): 120 | typeof fact outside: undefined
inferred names: anon arrowFn
strict: block function visible outside block? undefined
```

1. A **declaration** is created whole on scope entry, so it's callable above its line — letting a file's "story" sit on top with helpers below.
2. A `const` **expression** is in the TDZ until its line (Chapter 2 §2.2); a `var` **expression** holds `undefined` → "is not a function".

3. A **named function expression** (`function fact`) can call itself by name — for recursion (§4.10) and clearer **stack traces** (the list of nested calls printed with an error) — but the name isn't visible outside.
4. Anonymous functions assigned to variables get an **inferred** `name` (shown in stack traces).
5. In strict code, a function *declared inside a block* is block-scoped (sloppy mode has murky legacy rules — avoid it there).

Which to use?

Both are fine; be consistent. Common guidance: declarations (or `const` + arrow) for named top-level functions, arrows for callbacks; never rely on calling `const` functions before their definition.

TL;DR

Declarations are hoisted whole; expressions behave like their variable. Named function expressions can call themselves.

4.3 Arrow functions

What it is

Arrow functions (ES2015) are compact function expressions, `(params) => body`, that also differ in behaviour — chiefly, no own `this` (the object a function is working on — Chapter 9).

Syntax forms

Form	Example	Notes
one parameter	`n => n * n`	parentheses optional for exactly one simple parameter
several parameters	`(a, b) => a + b`	parentheses required
no parameters	`() => "hello"`	empty parentheses **required**
default / rest / destructured parameter	`(name = "Guest") => …`, `(...xs) => …`, `({ id }) => …`	parentheses required
expression body	`x => x * 2`	the expression's value is returned automatically (**implicit return**)
block body	`x => { const y = x * 2; return y; }`	braces → you must write `return`
returning an object literal (an object written directly in `{ }`)	`(x, y) => ({ x, y })`	wrap in parentheses, or the braces are read as a block

Example

```
// code/03_arrow_functions.js (abridged — the file first exercises every syntax form in the table)
"use strict"; // strict: a plain function call gets this === undefined
const makePointWrong = (x, y) => { x: x, y };    // parsed as a block with a label -> returns undefined
const makePoint = (x, y) => ({ x, y });

const timer = {
  seconds: 0,
  tickRegular() {
    return [1, 2, 3].map(function () { return this === undefined ? "undefined" : typeof this; });
  },
  tickArrow() {
    return [1, 2, 3].map(() => { this.seconds++; return this.seconds; });
  },
};

const counter = {
  count: 5,
  regular() { return this.count; },
  arrow: () => this?.count,                      // `this` comes from the surrounding (file-level) code
};

function outer() {
  const inner = () => arguments.length;          // uses outer's arguments
  return inner();
}
try { new square(2); } catch (e) { console.log("new on arrow ->", e.name + ":", e.message); }
```

Output (full run):

```
25 5 hello Hello, Guest! Hello, Ana! 2 word(s)
without (): undefined | with (): { x: 1, y: 2 }
with tax: [ 6, 14.4, 36 ] | over 10: [ 12, 30 ]
regular callback this: undefined | arrow callback: [ 1, 2, 3 ]
method regular: 5 | method arrow: undefined
arrow sees outer arguments.length: 3
new on arrow -> TypeError: square is not a constructor
```

How arrows differ from regular functions

Feature	Regular `function`	Arrow `=>`
`this`	set by **how** the function is called (Ch. 9)	**lexical**: taken from the surrounding code, fixed forever (even `call` / `bind` can't change it)
`arguments` object	yes	no — uses the enclosing function's, if any
Can be called with `new`	yes (declarations/expressions)	no — `TypeError`; no `prototype` property
`super` (reaches the parent class), `new.target` (tells whether `new` was used)	own	lexical
Can be a generator (a function that pauses and hands out values one at a time with `yield`, Ch. 8)	yes	no
Hoisted	declarations yes	no (it's an expression)
Good as object method	yes	**no** — `this` won't be the object
Good as callback inside a method	needs `bind` / `self = this`	**yes** — sees the method's `this`

In `tickRegular`, `map` calls the inner `function` plainly, so in strict mode its `this` is `undefined` — the object is lost. In `tickArrow` the arrow borrows `this` from the method (`timer`), so `this.seconds++` works — replacing the pre-ES2015 `var self = this;` / `.bind(this)` workarounds. Conversely `counter.arrow`, an arrow *as* a method, gets the `this` of where the literal was written: at the top of a CommonJS file (Node's original `require`/`module.exports` format) Node sets `this` to `module.exports` (an empty object), so `this.count` is `undefined`; in an ES module top-level `this` is `undefined`, same result.

Why it matters / benefits

- **Concise** callbacks: `items.filter(i => i.inStock).map(i => i.price)`.
- **Predictable** `this` in callbacks, timers and event handlers inside class methods.
- Ubiquitous in React (a UI library built from reusable *components*) and functional-style code.

When not to use arrows

- **Object methods** using `this` → method shorthand `name() { … }`.
- **Constructors** (called with `new`) → a `class` (Ch. 9).
- **DOM handlers needing `this` as the element** → regular function or `event.currentTarget` (Ch. 13).
- **Long multi-step logic** — a named declaration may read better.

In the interview

> **30-second answer — "Arrow vs regular functions?"** Arrows are shorter with implicit return for expression bodies, but the real difference: they don't bind their own `this`, `arguments`, `super` or `new.target` — they take them from the enclosing scope. Great as callbacks inside methods, wrong as object methods or constructors; no `new`, no `prototype`.

- *Change an arrow's `this` with `bind`?* No — `bind`/`call`/`apply` pass arguments but the `this` argument is ignored.
- *Return an object literal?* `() => ({ a: 1 })`.
- *All arguments in an arrow?* Rest parameter: `(...args) => args.length`.

TL;DR

`=> expression` returns it; `=> { … }` needs `return`. Lexical `this`, no `arguments`/`new` — perfect callbacks, poor methods.

4.4 Parameters: defaults, rest, `arguments` and spread

Default parameters — what problem they solve

Pre-ES2015 code patched missing arguments with `name = name || "Guest";` — extra code with a bug: `||` overwrites *every falsy value*, including deliberate `""` or `0`. **Default parameters** put the fallback in the **signature** (name + parameter list) — `function greet(name = "Guest")` — and replace **only** `undefined`.

How defaults work

1. Parameters initialise **left to right**.
2. If an argument is `undefined` (missing or explicit), the default is used; `null`, `0`, `""`, `false` are kept.
3. The default is an **expression evaluated at call time**, each time needed — it can call functions (`new Date()`, `nextId()`) and use **earlier** parameters.
4. Parameters behave like `let`: a default can't reference a *later* parameter (TDZ).

Example

```
// code/04_parameters.js (part 1, abridged — each output line labels what it shows)
function oldGreet(name) { name = name || "Guest"; return "Hello, " + name + "!"; }
function greet(name = "Guest") { return "Hello, " + name + "!"; }

function orderPizza(type = "Cheese", size = "Medium") {
  return `Ordering a ${size} ${type} pizza.`;
}
console.log(orderPizza(undefined, "Large"));   // skip the first by passing undefined

let counter = 0;
const nextId = () => ++counter;
function createUser(name, id = nextId(), tag = `${name.toLowerCase()}-${id}`) {
  return { name, id, tag };
}

function addItem(item, list = []) { list.push(item); return list; }

const clock = { now: () => "09:30:00" };
function logMessage(message, timestamp = clock.now()) { return `${timestamp}: ${message}`; }

function required(name) { throw new Error(`Missing parameter: ${name}`); }
function connect(url = required("url")) { return "connecting to " + url; }
```

Output (full run):

```
Hello, Guest! | Hello, Guest! | Hello, Ana!
empty string -> old: Hello, Guest! | new: "Hello, !"
undefined triggers the default: Hello, Guest! | null does not: Hello, null!
Ordering a Medium Cheese pizza.
Ordering a Medium Pepperoni pizza.
Ordering a Large Cheese pizza.
{ name: 'Ana', id: 1, tag: 'ana-1' } { name: 'Ben', id: 2, tag: 'ben-2' } { name: 'Cy', id: 99, tag: 'cy-99' }
[ 'a' ] [ 'b' ]
09:30:00: Server started | 10:00:00: Custom
Missing parameter: url
```

- **Several defaults**: to default an *earlier* parameter while supplying a later one, pass `undefined` — or use an options object.
- **Call-time evaluation**: `nextId()` ran per call; `tag` used earlier parameters. A real `timestamp = new Date().toLocaleTimeString()` gives each call its own time (a fixed clock keeps output reproducible).
- **Fresh default per call**: `list = []` is new each call — no Python-style shared mutable default.
- **Required parameters**: a throwing default turns a forgotten argument into a clear error.

Rest parameters, `arguments` and spread

```
// code/04_parameters.js (part 2)
function sum(...numbers) {
  let total = 0;
  for (const n of numbers) total += n;
  return total;
}
console.log("sum():", sum(), "| sum(1,2,3):", sum(1, 2, 3));
function tagAll(tag, ...words) { return words.map(w => `${tag}:${w}`); }
console.log(tagAll("js", "let", "const"));

function legacySum() {
  let total = 0;
  for (let i = 0; i < arguments.length; i++) total += arguments[i];
  return { total, isArray: Array.isArray(arguments), length: arguments.length };
}
console.log(legacySum(4, 5, 6));

const nums = [3, 9, 1];
console.log("Math.max(...nums):", Math.max(...nums), "| sum(...nums, 10):", sum(...nums, 10));
console.log("lengths:", orderPizza.length, sum.length, tagAll.length, createUser.length);

function makeButton({ label = "OK", color = "blue", disabled = false } = {}) {
  return `${label} (${color}${disabled ? ", disabled" : ""})`;
}
console.log(makeButton(), "|", makeButton({ label: "Save", disabled: true }));
```

Output:

```
sum(): 0 | sum(1,2,3): 6
[ 'js:let', 'js:const' ]
{ total: 15, isArray: false, length: 3 }
Math.max(...nums): 9 | sum(...nums, 10): 23
lengths: 0 0 1 1
OK (blue) | Save (blue, disabled)
```

Tool	Where	What it does
Rest parameter `...args`	last parameter in a definition	collects all remaining arguments into a **real array** (empty if none)
`arguments`	inside any non-arrow function	legacy **array-like** object of all arguments: has `length` and indexes, but no array methods; absent in arrows
Spread `...array`	in a call (or array/object literal)	expands an iterable (arrays, strings …) into separate arguments — the mirror image of rest
Options object with destructuring (pulling properties out into variables)	`function f({ a = 1, b = 2 } = {})`	named, order-independent parameters with defaults; the `= {}` lets you call `f()` with no argument at all
`fn.length`	property	number of parameters **before** the first one with a default or rest

Prefer rest over `arguments`: real arrays, work in arrows, self-describing. Destructuring is in Chapter 5; options objects are how most library APIs and React props (the settings object passed into a component) take many optional settings.

Common mistakes

- Expecting `null` to trigger a default — only `undefined` does; use `??` in the body for `null`.
- Required parameters after optional ones (`f(a = 1, b)`) — put defaults last or use an options object.
- In sloppy mode, mutating `arguments` also changes named parameters (and vice versa); in strict mode they're independent.

In the interview

- *When is a default evaluated?* At call time, each time the argument is `undefined`, left to right, with access to earlier parameters.
- *Rest vs spread?* Same `...`, opposite jobs: rest *collects* values into an array (parameter lists, destructuring); spread *expands* an iterable (calls, literals).

TL;DR

Defaults replace only `undefined`, per call, may use earlier parameters. `...rest` collects; spread expands; options objects give named parameters.

4.5 Scope: global, function and block

What it is

Scope is the region where a variable is visible. A name is looked up in the current scope, then each enclosing one outward.

Scope	Created by	Variables that live there	Visible
Global	top level of a script	top-level `var`, `let`, `const`, functions	everywhere
Function	every function body	`var`, `let`, `const` and parameters declared in the function	inside that function (and functions nested in it)
Block	any `{ … }`: `if`, `for`, `while`, `switch`, a bare block	`let`, `const`, `class` (and functions in strict mode)	only inside those braces

`var` **ignores blocks** (scoped to the nearest function or global); `let` / `const` **respect blocks**. ES modules add **module scope**: top-level names visible in that file only (Chapter 10). Inner scopes see outward; outer scopes can't see in.

Example

```
// code/05_scope.js (abridged — each output line labels what it shows)
function testScope() {
  if (true) {
    var functionScoped = "var: visible in the whole function";
    let blockLet = "let: only in this block";
  }
  console.log("after block:", functionScoped);
  try { blockLet; } catch (e) { console.log("after block, let ->", e.name + ":", e.message); }
}

const level = "global";
function showLevel() { return level; }
function caller() {
  const level = "caller";
  return showLevel();               // showLevel looks up `level` where it was DEFINED
}
console.log("lexical scope result:", caller());

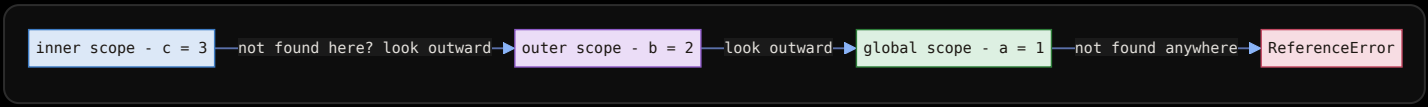
const a = 1;
function outer() {
  const b = 2;
  function inner() {
    const c = 3;
    return a + b + c;               // c local, b from outer, a from global
  }
  return inner();
}
console.log("scope chain sum:", outer());
```

Output (full run):

```
inside function: global var | global let | global const
inside block: let: only in this block / const: only in this block
after block: var: visible in the whole function
after block, let -> ReferenceError: blockLet is not defined
outside function, var -> ReferenceError
var loop counter after loop: 3 | let counter exists? undefined
shadowed: blue | outer still: red
lexical scope result: global
scope chain sum: 6
classic script globals -> var on global object: true | let: false | const: false
this CommonJS file's top-level var on globalThis? false
```

How it works — lexical scope and the scope chain

Lexical (static) scoping: visible scopes are decided by **where the function is written**, not where it's called. `showLevel` was written at top level, so `caller()` returns `"global"` despite its own `level`.



The **scope chain**: current scope → lexically enclosing scope → … → global → `ReferenceError: x is not defined` (or, when *assigning* in sloppy mode, an accidental global).

Shadowing: an inner declaration with the same name hides the outer one inside the inner scope (`paint`'s `color`); the outer is untouched. Legal but confusing — linters (e.g. ESLint) can flag it.

The global object and scripts vs modules

In a **classic browser script**, top-level `var` and function declarations become properties of the global object (`window` / `globalThis`); top-level `let` / `const` are global variables but not properties (the example shows this via Node's `vm` module). In **ES modules** and **Node CommonJS files**, top-level declarations are file-private — hence `globalVar` is not on `globalThis` in the last line.

Why it matters

- **Fewer collisions** — many functions can use `i` or `result`.
- **Encapsulation** — globals are shared by *every* script on the page, including third-party ads/analytics; keep the global scope clean.
- **Garbage collection** — once no code can reach a scope's variables, their memory can be freed.

Common mistakes

- `var` in an `if` / `for` visible after the block (`vi`).
- Expecting a variable from the *calling* function — scope is lexical, not dynamic.
- Forgetting `let` / `const` → accidental global (an error in strict mode, Chapter 1 §1.8).

In the interview

"Explain scope in JavaScript." Scope decides where a name is visible: global, function, block (plus module). `var` is function-scoped, `let` / `const` block-scoped. Scoping is lexical: a function sees the variables where it's *defined*, walking outward along the scope chain — which is what makes closures possible.

- *Shadowing?* An inner declaration hiding a same-named outer variable within the inner scope.
- *Does top-level `let` create a `window` property?* No; only top-level `var` and function declarations in classic scripts.

TL;DR

Global → function → block. `var` function-scoped, `let` / `const` block-scoped. Lookups walk the lexical scope chain; unresolved names throw `ReferenceError`.

4.6 Hoisting — the complete picture

Extending Chapter 2 §2.2: every declaration kind observed *before* its line runs.

```js
// code/06_hoisting.js (the report part)
function demo() {
  const report = [];
  report.push(["function declaration", typeof hoistedFn === "function" ? hoistedFn() : "missing"]);
  report.push(["var", String(v)]);
  try { l; } catch (e) { report.push(["let", e.name]); }
  try { c; } catch (e) { report.push(["const", e.name]); }
  try { new K(); } catch (e) { report.push(["class", e.name]); }
  try { fe(); } catch (e) { report.push(["var function expression", e.name + " (" + typeof fe + ")"]); }

  function hoistedFn() { return "callable"; }
  var v = 1;
  let l = 2;
  const c = 3;
  class K {}
  var fe = function () {};
  return report;
}
```

Output (full run):

```
function declaration     -> callable
var                      -> undefined
let                      -> ReferenceError
const                    -> ReferenceError
class                    -> ReferenceError
var function expression  -> TypeError (undefined)
ReferenceError: inner x is in its TDZ
var shadow hoisting: [ undefined, 'grumpy' ]
```

Declaration	Hoisted?	Value before its line
`function f() {}`	yes, with its body	the function (callable)
`var x`	yes	`undefined`
`let x`, `const x`	yes	uninitialised → `ReferenceError` (TDZ)
`class X {}`	yes	uninitialised → `ReferenceError` (TDZ)
`var f = function () {}` / `const f = () => {}`	as the variable	`undefined` (→ `TypeError` if called) / TDZ

The last two output lines:

- `let` **shadowing with TDZ**: a function with a later `let x` can't read an *outer* `x` above that line — the inner `x` already exists (in its TDZ) for the whole block.
- `var` **shadowing**: in `moodSwing`, `mood` reads `undefined` instead of the global `"happy"` because the function's own `var mood` is hoisted.

Mental model: the engine first registers every declaration in a scope, then runs the code. Nothing physically moves.

4.7 Closures

What it is

A **closure** is a function together with the variables of the scope where it was created. Through lexical scoping an inner function uses outer variables — and **keeps using them after the outer function has returned**; they stay alive as long as the inner function exists. Think of a backpack of (references to) surrounding variables the function carries everywhere.

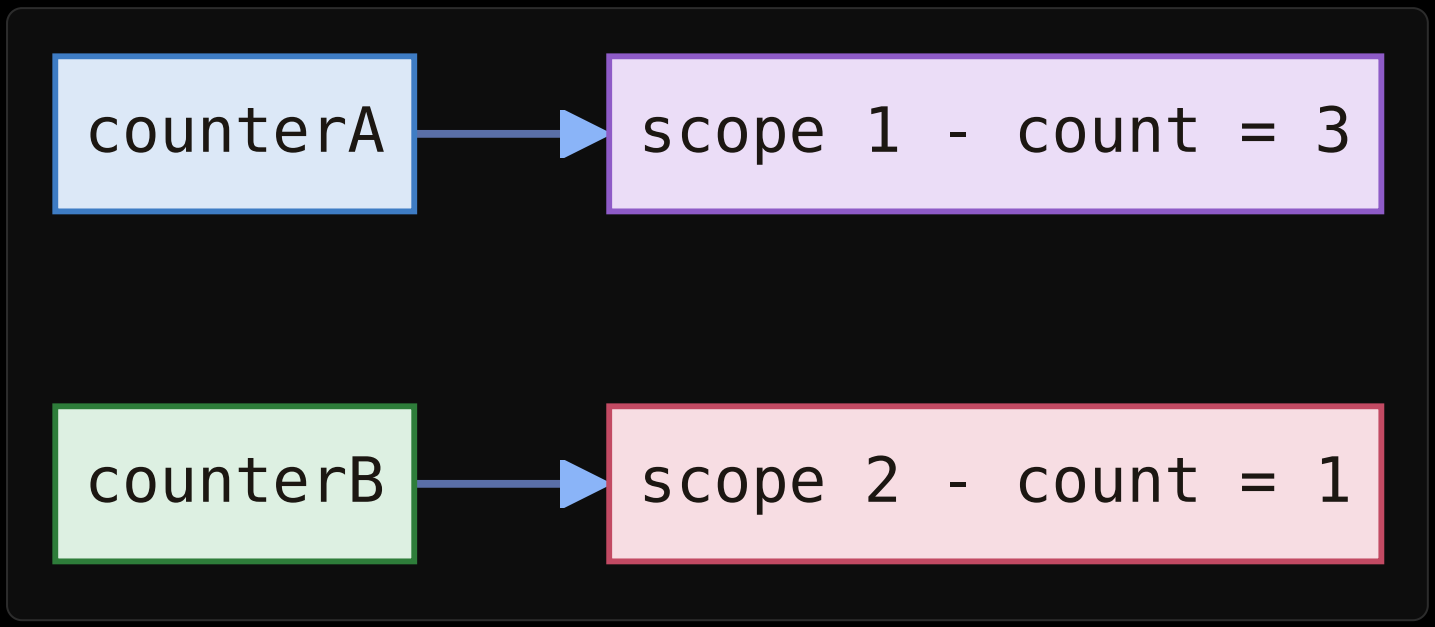
Example — a counter with private state

```
// code/07_closures.js (part 1)
function makeCounter() {
  let count = 0;                        // private: only the returned function can reach it
  return function () {
    count++;
    return count;
  };
}
const counterA = makeCounter();
const counterB = makeCounter();         // an independent closure with its own count
console.log("A:", counterA(), counterA(), counterA(), "| B:", counterB());
```

Output:

```
A: 1 2 3 | B: 1
```

1. `makeCounter()` creates a fresh scope with `count = 0`.
2. The inner function captures a reference to that scope and is returned.
3. `makeCounter` ends, but the scope survives because the function references it.
4. Each `counterA()` increments *that* `count`; nothing else can touch it.
5. Another `makeCounter()` call creates a **new** scope — `counterB` is independent.



The module pattern — several functions sharing private state

```
// code/07_closures.js (part 2)
function createBankAccount(initial) {
  let balance = initial;
  const history = [];
  function record(type, amount) { history.push(`${type} ${amount}`); }
  return {
    deposit(amount) { balance += amount; record("deposit", amount); return balance; },
    withdraw(amount) {
      if (amount > balance) throw new Error("Insufficient funds");
      balance -= amount; record("withdraw", amount); return balance;
    },
    get balance() { return balance; },
    history: () => [...history],
  };
}
const acct = createBankAccount(100);
acct.deposit(50);
acct.withdraw(30);
console.log("balance:", acct.balance, "| direct access:", acct.balanceVar, "| history:", acct.history());
try { acct.withdraw(1000); } catch (e) { console.log("withdraw 1000 ->", e.message); }
```

Output:

```
balance: 120 | direct access: undefined | history: [ 'deposit 50', 'withdraw 30' ]
withdraw 1000 -> Insufficient funds
```

All methods close over the same `balance`/`history`; the balance changes only through rule-enforcing `deposit`/`withdraw`, and `history()` returns a copy. This is **encapsulation** without classes — the pre-ES2015 **module pattern** (§4.9). Classes now offer `#private` fields (Chapter 9).

More closure patterns

```
// code/07_closures.js (part 3)
function makeMultiplier(factor) { return (x) => x * factor; }
const double = makeMultiplier(2), triple = makeMultiplier(3);
console.log("double(5):", double(5), "| triple(5):", triple(5));

function makeReader() {
  let message = "first";
  const read = () => message;
  message = "changed later";
  return read;
}
console.log("closure sees latest value:", makeReader()());

function once(fn) {
  let done = false, result;
  return (...args) => {
    if (!done) { done = true; result = fn(...args); }
    return result;
  };
}
const init = once(() => { console.log("  initialising..."); return 42; });
console.log("init:", init(), init(), init());

function memoize(fn) {
  const cache = new Map();
  return (n) => {
    if (cache.has(n)) return cache.get(n);
    const value = fn(n);
    cache.set(n, value);
    return value;
  };
}
let slowCalls = 0;
const slowSquare = (n) => { slowCalls++; return n * n; };
const fastSquare = memoize(slowSquare);
fastSquare(9); fastSquare(9); fastSquare(9);
console.log("memoized: underlying function ran", slowCalls, "time(s)");
```

Output:

```
double(5): 10 | triple(5): 15
closure sees latest value: changed later
  initialising...
init: 42 42 42
memoized: underlying function ran 1 time(s)
```

- **Function factories** — configure once (`factor`), get a specialised function.
- **Closures capture variables, not values** — `read` sees the value at *call* time.
- `once` — `done` flag in the closure; one-time init, "submit only once" buttons.
- `memoize` (**memoization** = remembering results of earlier calls) — the **cache** lives in the closure, so repeated inputs skip the work. (Real memoizers must handle cache size and multi-argument keys.)

The classic loop puzzle

```
// code/07_closures.js (part 4)
const withVar = [], withLet = [];
for (var i = 0; i < 3; i++) withVar.push(() => i);
for (let j = 0; j < 3; j++) withLet.push(() => j);
console.log("var closures:", withVar.map(f => f()), "| let closures:", withLet.map(f => f()));

for (var t = 0; t < 3; t++) setTimeout(() => process.stdout.write(`var-timer ${t}  `), 0);
for (let u = 0; u < 3; u++) setTimeout(() => process.stdout.write(`let-timer ${u}  `), 0);
setTimeout(() => console.log(), 0);
```

Output:

```
var closures: [ 3, 3, 3 ] | let closures: [ 0, 1, 2 ]
var-timer 3  var-timer 3  var-timer 3  let-timer 0  let-timer 1  let-timer 2
```

With `var` there is **one** function-scoped `i`; all closures share it, and when they run the loop has finished (`i` is 3). A `let` in a `for` header creates a **new binding** (fresh name-to-value slot) **per iteration**, copying the current value, so each closure has its own `j`. The timers are the same story with async callbacks (Chapter 12). Pre-ES2015 fix: an IIFE per iteration (§4.9); today, `let`.

Why closures matter — real-world examples

- **Event handlers** remember their data: `addEventListener("click", () => deleteItem(item.id))` (Ch. 13).
- **Debounce / throttle** (debounce = run once after calls stop for a moment; throttle = at most once per time window) keep timer IDs in a closure (Ch. 17).
- **React hooks**: each render's handlers close over that render's props and state — the source of "stale closure" bugs.
- **Callbacks with context**: `setTimeout(() => notify(user), 1000)` keeps `user` alive.
- **Private state** and factories.

Common mistakes and costs

- **Stale closures** — reading a variable from an old scope (an old React render) when you expected the latest.
- **Memory** — anything a live closure references can't be garbage-collected; a long-lived event listener capturing a huge array keeps it alive. Remove unneeded listeners.
- **The loop puzzle** — use `let`.

In the interview

> **30-second answer — "What is a closure?"** A function bundled with references to the variables of the scope where it was defined. Because scope is lexical, an inner function accesses outer variables — and keeps that access after the outer function returns, so they stay alive. Closures give private state (counters, module pattern), factories, memoization, and underpin event handlers and callbacks.

- *`for (var i…) setTimeout(() => log(i))` prints the final value thrice?* One shared `i`; callbacks run after the loop. `let` gives one binding per iteration.
- *Do closures copy values?* No — live references; later changes show up.
- *Memory leaks (memory never freed because something still points at it)?* Long-lived closures (listeners, timers, caches) holding large objects keep them reachable.

Self-check

1. `const c = makeCounter(); c(); c();` — next `c()`? → `3`.
2. Do `makeCounter()` calls share a count? → No; new scope each call.
3. Variable only two functions can modify? → Declare it in an outer function, return both (module pattern).

TL;DR

Closure = function + the variables around its definition, alive as long as the function. Private state, factories, `once`/`memoize`; `let` in loops; beware stale values and retained memory.

4.8 Higher-order functions and callbacks

What it is

Because functions are **first-class**:

- A **callback** is a function passed to another function so it can call it ("call back") at the right moment.
- A **higher-order function** takes a function as argument, returns one, or both.

Example

```
// code/08_higher_order_callbacks.js (abridged — each output line labels what it shows)
const paidTotals = orders.filter((o) => o.status === "paid").map((o) => o.total);
const revenue = paidTotals.reduce((sum, t) => sum + t, 0);

function greaterThan(limit) { return (n) => n > limit; }      // returns a function

const compose = (...fns) => (x) => fns.reduceRight((acc, fn) => fn(acc), x);
const pipe = (...fns) => (x) => fns.reduce((acc, fn) => fn(acc), x);
const slugify = pipe(trim, lower, dashes);                     // trim, then lower, then dashes

const curriedAdd = (a) => (b) => (c) => a + b + c;
const baseTen = volume.bind(null, 10);                         // partial application

console.log("A: before");
[1].forEach(() => console.log("B: forEach callback runs immediately (synchronous)"));
setTimeout(() => console.log("D: setTimeout callback runs later (asynchronous)"), 0);
console.log("C: after");

function hits() { hits.count = (hits.count ?? 0) + 1; }      // functions can carry properties
```

Output (full run):

```
callback call 0
callback call 1
callback call 2
paid totals: [ 25, 40 ] | revenue: 65
sorted by total desc: [ 2, 3, 1 ]
[5, 15, 25] > 10: [ 15, 25 ]
slug: hello-big-world | compose order: a-b
curried: 6 20
partial: 60
A: before
B: forEach callback runs immediately (synchronous)
C: after
function property count: 2
D: setTimeout callback runs later (asynchronous)
```

The patterns

Pattern	Idea	Example above
Callback	pass behaviour in	`repeat(3, action)`
Array HOFs (higher-order functions)	`map`, `filter`, `reduce`, `sort`, `find`, `some` … take callbacks (Ch. 7)	revenue of paid orders
Returning a function	build a predicate (true/false function)/handler from settings	`greaterThan(10)`
Composition	combine small functions into a pipeline; `pipe` runs left→right, `compose` right→left	`slugify`
Currying	turn `f(a, b, c)` into `f(a)(b)(c)` so arguments can be supplied one at a time	`curriedAdd`
Partial application	fix some arguments now, supply the rest later (`bind` can do it)	`baseTen`

Synchronous vs asynchronous callbacks

Order **A, B, C … D**: `forEach` calls its callback *immediately* (**synchronous**); `setTimeout` only *registers* it to run after the current code finishes (**asynchronous**). Event listeners, timers and network responses use asynchronous callbacks, scheduled by the event loop (Chapter 12, which also covers "callback hell" and promises). The last line shows functions are objects that can carry properties (hence `.name`, `.length`, `.bind`).

Why it matters

- Separates **what** from **how/when**: `filter` loops; you supply the rule.
- **Declarative** (describe *what*, not each step), reusable code — the base of functional programming (small functions returning new values instead of mutating) and every event-driven API.

When not to use

- Long chains of tiny anonymous callbacks hurt readability and debugging — name them.
- In **hot code** (runs very often) a plain loop can be marginally faster than chained `map/filter` — measure first.

TL;DR

Callbacks are passed-in functions; higher-order functions take or return functions — enabling array methods, composition, currying and every event/async API.

4.9 IIFE — Immediately Invoked Function Expressions

What it is

An **IIFE** ("iffy") is a function expression called where it's defined: `(function () { … })();` or `(() => { … })();`. The parentheses make `function` an *expression*; a statement starting with `function` is a declaration and can't be called immediately (`SyntaxError`).

The problem it solves

Before `let`/`const` and modules, a function was the only way to create private scope. Wrapping a script in an IIFE kept variables out of the global scope; an IIFE returning an object of public functions was *the* **module pattern** of the 2000s (jQuery — then the dominant web-page library — plugins, early libraries).

Example

```
// code/09_iife.js
(function () {
  const secret = "only visible in here";
  console.log("IIFE ran:", secret);
})();
console.log("secret outside?", typeof secret);

const config = ((env) => {
  const base = { retries: 3 };
  return env === "prod" ? { ...base, debug: false } : { ...base, debug: true };
})("dev");
console.log("config:", config);

const Cart = (function () {
  const items = [];                          // private
  function total() { return items.reduce((s, i) => s + i.price, 0); }
  return {
    add(name, price) { items.push({ name, price }); return this; },
    total,
    count: () => items.length,
  };
})();
Cart.add("pen", 2).add("book", 12);
console.log("Cart:", Cart.count(), "items, total", Cart.total(), "| items exposed?", "items" in Cart);

(async () => {
  const value = await Promise.resolve("async IIFE done");
  console.log(value);
})();

try {
  eval("function () {}()");
} catch (e) {
  console.log("function statement called directly ->", e.name);
}
```

Output:

```
IIFE ran: only visible in here
secret outside? undefined
config: { retries: 3, debug: true }
Cart: 2 items, total 14 | items exposed? false
function statement called directly -> SyntaxError
async IIFE done
```

(The async IIFE prints last because `await` always resumes asynchronously — Chapter 12.)

Today

Old need	Modern replacement
private scope for a script	ES modules (each file has its own scope) or a plain `{ }` block with `let` / `const`
module pattern	ES modules with `export` (Ch. 10) or classes with `#private`
capture loop variable	`let` in the loop header
`await` at the top level	top-level `await` in modules

IIFEs still appear in bundler output, in snippets that must not leak globals (analytics tags), and as async IIFEs in CommonJS scripts. If a line starts with `(` and the previous lacks a semicolon, ASI (Automatic Semicolon Insertion) won't insert one — hence `;(function () { … })()`.

TL;DR

`(function () { … })()` creates and runs a private scope; modules and block scope have mostly replaced it.

4.10 Recursion

What it is

A **recursive** function calls itself on a smaller version of the problem. It needs a **base case** (answered directly, stops recursion) and a **recursive case** (calls itself on a *smaller* input, moving toward the base case).

Example

```
// code/10_recursion.js (abridged — the file also has totalSize over a folder tree, flatten, and fibLoop)
function factorial(n) {
  if (n <= 1) return 1;          // base case: stops the recursion
  return n * factorial(n - 1);   // recursive case: smaller input
}

function countdown(n, depth = 0) {
  const pad = "  ".repeat(depth);
  if (n === 0) { console.log(pad + "liftoff (base case)"); return; }
  console.log(pad + "countdown(" + n + ")");
  countdown(n - 1, depth + 1);
  console.log(pad + "back in countdown(" + n + ")");
}

function forever(n) { return forever(n + 1); }              // no base case

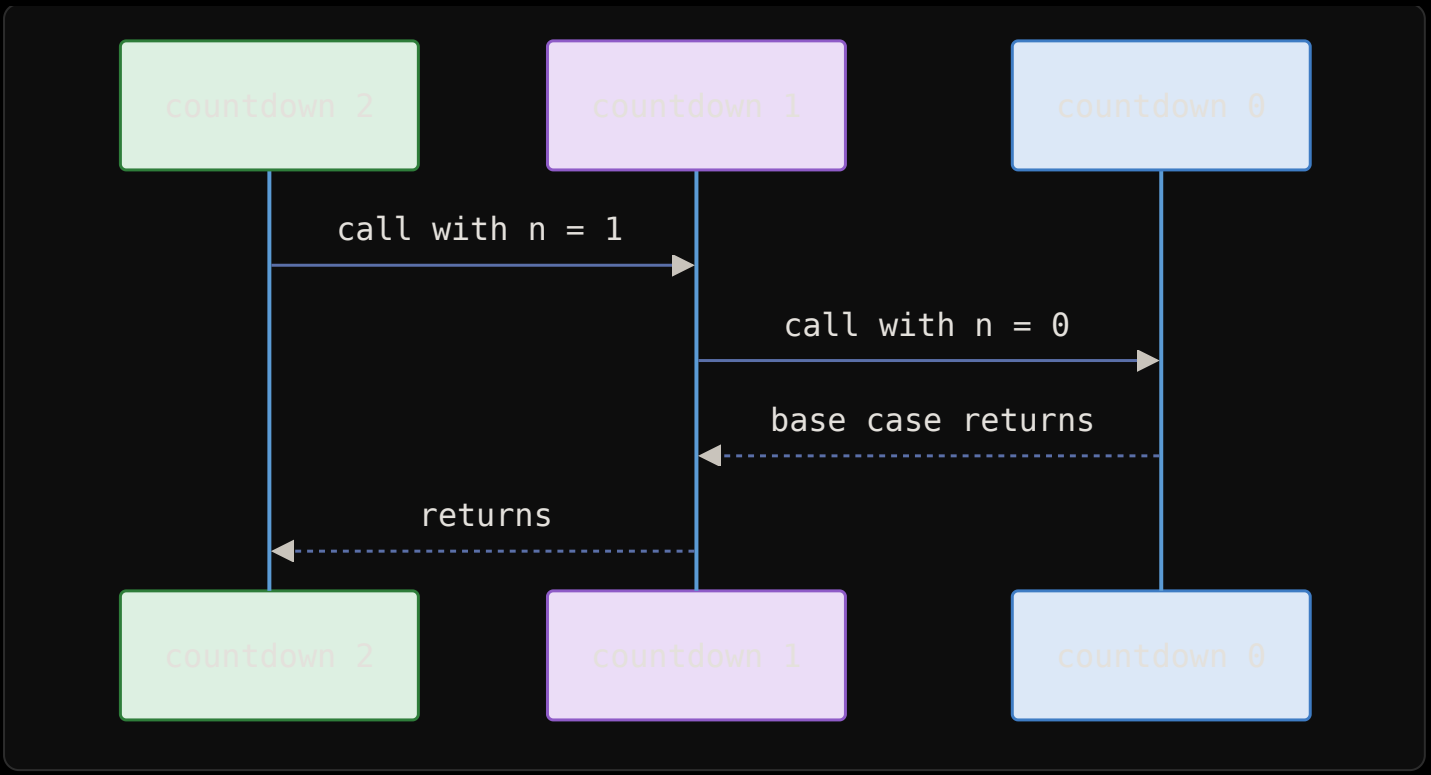
let calls = 0;
function fib(n) { calls++; return n < 2 ? n : fib(n - 1) + fib(n - 2); }
```

Output (full run):

```
factorial(5): 120
countdown(2)
  countdown(1)
    liftoff (base case)
  back in countdown(1)
back in countdown(2)
total size: 15
flatten: [ 1, 2, 3, 4, 5 ]
no base case -> RangeError: Maximum call stack size exceeded
naive fib(25) calls: 242785
fibLoop(25): 75025 | fibLoop(50): 12586269025
```

How it works — the call stack

Each call gets its own execution context on the **call stack**: calls go *down* (each waiting), the base case returns, then waiting calls resume in reverse order.



The stack is limited (roughly ten thousand **frames** — one per active call — depending on engine and frame size). Too deep, or no base case, throws `RangeError: Maximum call stack size exceeded` ("stack overflow"). Engines other than Safari's do **not** implement **tail-call optimisation** (reusing the frame when the last action is a self-call), so tail-call form doesn't make deep recursion safe.

When to use it (and when not)

Good fit	Better as a loop
tree-shaped data: file systems, the DOM, JSON, org charts, comments with replies	simple counting / summing over a flat list
divide-and-conquer algorithms (split the problem into halves, solve each, combine): merge sort, quicksort, binary search	inputs that could nest tens of thousands of calls deep
backtracking (try a choice, undo it if it leads nowhere): permutations, sudoku	naive recursions that repeat work (Fibonacci) unless memoized

Naive `fib(25)` made **242,785** calls by recomputing values — exponential time. Memoization (§4.7) or a loop makes it linear. `fibLoop` stops at 50 on purpose: `fib(79)` and up exceed `Number.MAX_SAFE_INTEGER` and silently lose precision (Chapter 2 §2.10) — use `BigInt`.

TL;DR

Base case + smaller recursive case. Natural for trees and divide-and-conquer; watch stack depth (`RangeError`) and repeated work (memoize).

4.11 Pure functions

What it is

A **pure function** (1) returns the **same output for the same inputs** and (2) has **no side effects** — no mutating arguments or globals, no I/O (input/output: files, keyboard, network), no DOM changes, no logging. Otherwise it's **impure**: `Math.max` is pure; `Math.random`, `Date.now`, `console.log`, `array.push` are not.

Example

```
// code/11_pure_functions.js (abridged)
const addTax = (price, rate) => +(price * (1 + rate)).toFixed(2);   // pure
console.log("pure:", addTax(100, 0.2), addTax(100, 0.2));

let rate = 0.2;
const addTaxImpure = (price) => +(price * (1 + rate)).toFixed(2);   // reads outside state
console.log("impure before:", addTaxImpure(100));
rate = 0.25;
console.log("impure after someone changed rate:", addTaxImpure(100));

function addItemMutating(cart, item) { cart.push(item); return cart; }  // side effect
function addItemPure(cart, item) { return [...cart, item]; }          // returns a new array
const cart = ["apple"];
const newCart = addItemPure(cart, "pear");
console.log("pure add -> original:", cart, "| new:", newCart);
addItemMutating(cart, "plum");
console.log("mutating add -> original changed:", cart);

const assert = require("node:assert");
assert.strictEqual(addTax(10, 0.1), 11);
assert.deepStrictEqual(addItemPure([], "x"), ["x"]);
console.log("tests passed");
```

Output:

```
pure: 120 120
impure before: 120
impure after someone changed rate: 125
pure add -> original: [ 'apple' ] | new: [ 'apple', 'pear' ]
mutating add -> original changed: [ 'apple', 'plum' ]
Date.now() is impure: number
tests passed
```

Why it matters

- **Testable** — no setup or **mocks** (fake stand-ins for dependencies): call and compare.
- **Safe to cache** (memoize) and run in any order or in parallel.
- **Frameworks rely on it**: React components must be pure functions of props and state while rendering; Redux **reducers** (compute next state from current state + action) must be pure. Returning a new array (`[...cart, item]`) lets them detect changes by reference (Chapter 2 §2.4).

Trade-offs

A program with no side effects does nothing visible. Aim for a **pure core, impure shell**: pure calculations, side effects at the edges (event handlers, API calls). Copying instead of mutating costs some memory/time — negligible for most UI-sized data.

TL;DR

Same inputs → same output, no side effects. Keep business logic (app-specific rules like prices and discounts) pure; isolate I/O and mutation.

Chapter 4 TL;DR

- **Functions** are first-class values: missing arguments and missing `return` give `undefined`; objects are passed as shared references.
- **Declarations** are hoisted whole; **expressions** (incl. arrows) follow their variable (`const` → TDZ).
- **Arrows**: concise (`x => x * 2`, `() => ({…})`), lexical `this`, no `arguments`, no `new` — ideal callbacks, wrong as methods.

- **Parameters**: defaults replace only `undefined`, per call, may use earlier params; `...rest` collects; spread expands; options objects give named parameters.
- **Scope**: global, function (`var`), block (`let`/`const`); lookups walk the **lexical** scope chain; inner names shadow outer.
- **Closures**: live access to the variables where a function was defined — private state, factories, `once`, `memoize`, handlers. `let` in loops = one binding per iteration.
- **Higher-order functions & callbacks**: `map/filter/reduce`, composition, currying; callbacks sync or async.
- **IIFE**: `(() => { … })()` — legacy private scope/module pattern, replaced by modules.
- **Recursion**: base case + smaller case; beware `RangeError` and repeated work.
- **Pure functions**: same input → same output, no side effects — easy to test, required by React and Redux.

Next: Chapter 5 — Objects, Math, Date and JSON. For `this`, `call/apply/bind` and classes see Chapter 9; for `try/catch`, Chapter 11.